Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.78)

Metric

$$ds^2 = \left(-N\left(y\right)^{\frac{1}{\alpha}-1} Q\left(z\right)^{-\frac{1}{\alpha}-1} A\left(t\right)\right) \mathrm{d}t \otimes \mathrm{d}t + \left(N\left(y\right)^{\frac{1}{\alpha}-1} Q\left(z\right)^{-\frac{1}{\alpha}-1} A\left(t\right)\right) \mathrm{d}x \otimes \mathrm{d}x + \left(t^{\alpha+1} N\left(y\right)^{\frac{1}{\alpha}-1} Q\left(z\right)^{-\frac{1}{\alpha}-1}\right) \mathrm{d}y \otimes \mathrm{d}y + \left(t^{-\alpha+1} N\left(y\right)^{\frac{1}{\alpha}-1} Q\left(z\right)^{-\frac{1}{\alpha}-1}\right) \mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$$(x^{\mu}) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{\frac{1}{2}} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}} \sqrt{A\left(t\right)}}{\sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dt + \left(\frac{\sqrt{\frac{1}{2}} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}} \sqrt{A\left(t\right)}}{\sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dx$$

$$n = \left(\frac{\sqrt{\frac{1}{2}} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}} \sqrt{A\left(t\right)}}{\sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dt + \left(-\frac{\sqrt{\frac{1}{2}} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}} \sqrt{A\left(t\right)}}{\sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dx$$

$$m = \left(\frac{\sqrt{\frac{1}{2}} \sqrt{t} t^{\frac{1}{2}\alpha} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}}}{\sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dy + \left(\frac{i \sqrt{\frac{1}{2}} \sqrt{t} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}}}{t^{\frac{1}{2}\alpha} \sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dz$$

$$\overline{m} = \left(\frac{\sqrt{\frac{1}{2}} \sqrt{t} t^{\frac{1}{2}\alpha} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}}}{\sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dy + \left(-\frac{i \sqrt{\frac{1}{2}} \sqrt{t} N\left(y\right)^{\frac{1}{2\alpha}} Q\left(z\right)^{-\frac{1}{2\alpha}}}{t^{\frac{1}{2}\alpha} \sqrt{N\left(y\right)} \sqrt{Q\left(z\right)}}\right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\right)dt + \left(\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\right)dx$$

$$n = \left(-\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\right)dt + \left(-\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\right)dx$$

$$m = \left(\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}t^{\frac{1}{2}\alpha}}\right)dy + \left(\frac{i\sqrt{2}t^{\frac{1}{2}\alpha}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}}\right)dz$$

$$\overline{m} = \left(\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}t^{\frac{1}{2}\alpha}}\right)dy + \left(-\frac{i\sqrt{2}t^{\frac{1}{2}\alpha}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}}\right)dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\partial_t + \frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\partial_x$$

$$\Delta = -\frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\partial_t - \frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{A\left(t\right)}}\partial_x$$

$$\delta = \frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}t^{\frac{1}{2}\alpha}}\partial_y + \frac{i\sqrt{2}t^{\frac{1}{2}\alpha}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}}\partial_z$$

$$\overline{\delta} = \frac{\sqrt{2}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}t^{\frac{1}{2}\alpha}}\partial_y - \frac{i\sqrt{2}t^{\frac{1}{2}\alpha}N\left(y\right)^{-\frac{1}{2\alpha}}Q\left(z\right)^{\frac{1}{2\alpha}}\sqrt{N\left(y\right)}\sqrt{Q\left(z\right)}}{2\sqrt{t}}\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \sqrt{Q(z)}}{4\, t \sqrt{A(t)}}$$

$$\sigma = \frac{\sqrt{2} \alpha N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \sqrt{Q(z)}}{4\, t \sqrt{A(t)}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \sqrt{Q(z)}}{4\, t \sqrt{A(t)}}$$

$$\lambda = -\frac{\sqrt{2} \alpha N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \sqrt{Q(z)}}{4\, t \sqrt{A(t)}}$$

$$\nu = 0$$

$$\alpha = -\frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{8 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} + \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{8 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}} - \frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{8\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} - \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{8\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}}$$

$$\beta = -\frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{8 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} - \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{8 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}} - \frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{8\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} + \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{8\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}}$$

$$\tau = \frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{4 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} + \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{4 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}} + \frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{4\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} - \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{4\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}}$$

$$\pi = \frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{4 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} - \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{4 \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}} + \frac{i \sqrt{2} t^{\alpha} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{N(y)} \frac{\partial}{\partial z} Q(z)}{4\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{Q(z)}} + \frac{\sqrt{2} N(y)^{-\frac{1}{2\alpha}} Q(z)^{\frac{1}{2\alpha}} \sqrt{Q(z)} \frac{\partial}{\partial y} N(y)}{4\, \alpha \sqrt{t} t^{\frac{1}{2}\alpha} \sqrt{N(y)}}$$

$$\epsilon = -\frac{\sqrt{2}N(y)^{-\frac{1}{2\alpha}}Q(z)^{\frac{1}{2\alpha}}\sqrt{N(y)}\sqrt{Q(z)}\frac{\partial}{\partial t}A(t)}{8A(t)^{\frac{3}{2}}}$$

$$\gamma = \frac{\sqrt{2}N(y)^{-\frac{1}{2\alpha}}Q(z)^{\frac{1}{2\alpha}}\sqrt{N(y)}\sqrt{Q(z)}\frac{\partial}{\partial t}A(t)}{8A(t)^{\frac{3}{2}}}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\alpha^2 N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)}{8t^2A(t)} + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)\frac{\partial}{\partial t}A(t)}{4tA(t)^2} + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)}{8t^2A(t)}$$

$$\Phi_{01} = -\frac{i\alpha t^{\alpha}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)}{8t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}} + \frac{\alpha Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)}{8t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}} + \frac{it^{\alpha}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)}{8\alpha t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}} - \frac{Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)}{8\alpha t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}}$$

$$\begin{aligned}\Phi_{02} = {} & \frac{iN(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}\frac{\partial}{\partial y}N(y)\frac{\partial}{\partial z}Q(z)}{4t} + \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)^2}{8tt^{\alpha}Q(z)} - \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial^2}{(\partial z)^2}Q(z)}{4tt^{\alpha}} - \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)^2}{8tt^{\alpha}N(y)} \\ & + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial^2}{(\partial y)^2}N(y)}{4tt^{\alpha}} - \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial^2}{(\partial z)^2}Q(z)}{4\alpha tt^{\alpha}} - \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial^2}{(\partial y)^2}N(y)}{4\alpha tt^{\alpha}} \\ & - \frac{iN(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}\frac{\partial}{\partial y}N(y)\frac{\partial}{\partial z}Q(z)}{4\alpha^2 t} - \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)^2}{8\alpha^2 tt^{\alpha}Q(z)} + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)^2}{8\alpha^2 tt^{\alpha}N(y)}\end{aligned}$$

$$\Phi_{10} = \frac{i\alpha t^{\alpha}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)}{8t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}} + \frac{\alpha Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)}{8t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}} - \frac{it^{\alpha}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)}{8\alpha t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}} - \frac{Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)}{8\alpha t^{\frac{3}{2}}t^{\frac{1}{2}\alpha}N(y)^{\left(\frac{1}{\alpha}\right)}\sqrt{A(t)}}$$

$$\begin{aligned}\Phi_{11} = {} & -\frac{\alpha^2 N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)}{16t^2A(t)} + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)\frac{\partial}{\partial t}A(t)^2}{8A(t)^3} - \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)\frac{\partial^2}{(\partial t)^2}A(t)}{8A(t)^2} - \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)^2}{16tt^{\alpha}Q(z)} \\ & + \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial^2}{(\partial z)^2}Q(z)}{8tt^{\alpha}} - \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)^2}{16tt^{\alpha}N(y)} + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial^2}{(\partial y)^2}N(y)}{8tt^{\alpha}} + \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial^2}{(\partial z)^2}Q(z)}{8\alpha tt^{\alpha}} \\ & + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)Q(z)}{16t^2A(t)} - \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial^2}{(\partial y)^2}N(y)}{8\alpha tt^{\alpha}} + \frac{t^{2\alpha}N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}N(y)\frac{\partial}{\partial z}Q(z)^2}{16\alpha^2 tt^{\alpha}Q(z)} + \frac{N(y)^{-\frac{1}{\alpha}}Q(z)^{\left(\frac{1}{\alpha}\right)}Q(z)\frac{\partial}{\partial y}N(y)^2}{16\alpha^2 tt^{\alpha}N(y)}\end{aligned}$$

$$\Phi_{12} = -\frac{i\,\alpha t^{\alpha} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)}{8\,t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}} + \frac{\alpha Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)}{8\,t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}} + \frac{i\,t^{\alpha} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)}{8\,\alpha t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}} - \frac{Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)}{8\,\alpha t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}}$$

$$\begin{aligned}\Phi_{20} = &-\frac{i\,N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} \frac{\partial}{\partial y} N(y) \frac{\partial}{\partial z} Q(z)}{4\,t} + \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)^2}{8\,tt^{\alpha} Q(z)} - \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial^2}{(\partial z)^2} Q(z)}{4\,tt^{\alpha}} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)^2}{8\,tt^{\alpha} N(y)} \\ &+ \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial^2}{(\partial y)^2} N(y)}{4\,tt^{\alpha}} - \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial^2}{(\partial z)^2} Q(z)}{4\,\alpha tt^{\alpha}} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial^2}{(\partial y)^2} N(y)}{4\,\alpha tt^{\alpha}} \\ &+ \frac{i\,N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} \frac{\partial}{\partial y} N(y) \frac{\partial}{\partial z} Q(z)}{4\,\alpha^2 t} - \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)^2}{8\,\alpha^2 tt^{\alpha} Q(z)} + \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)^2}{8\,\alpha^2 tt^{\alpha} N(y)}\end{aligned}$$

$$\Phi_{21} = \frac{i\,\alpha t^{\alpha} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)}{8\,t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}} + \frac{\alpha Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)}{8\,t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}} - \frac{i\,t^{\alpha} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)}{8\,\alpha t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}} - \frac{Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)}{8\,\alpha t^{\frac{3}{2}} t^{\frac{1}{2}\,\alpha} N(y)^{\left(\frac{1}{\alpha}\right)} \sqrt{A(t)}}$$

$$\Phi_{22} = -\frac{\alpha^2 N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z)}{8\,t^2 A(t)} + \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial}{\partial t} A(t)}{4\,tA(t)^2} + \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z)}{8\,t^2 A(t)}$$

$$\begin{aligned}\Lambda = &\frac{\alpha^2 N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z)}{48\,t^2 A(t)} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial}{\partial t} A(t)^2}{24\,A(t)^3} + \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial^2}{(\partial t)^2} A(t)}{24\,A(t)^2} \\ &- \frac{3\,t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)^2}{16\,tt^{\alpha} Q(z)} + \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial^2}{(\partial z)^2} Q(z)}{8\,tt^{\alpha}} - \frac{3\,N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)^2}{16\,tt^{\alpha} N(y)} \\ &+ \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial^2}{(\partial y)^2} N(y)}{8\,tt^{\alpha}} - \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)^2}{4\,\alpha tt^{\alpha} Q(z)} + \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial^2}{(\partial z)^2} Q(z)}{8\,\alpha tt^{\alpha}} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z)}{48\,t^2 A(t)} \\ &+ \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)^2}{4\,\alpha tt^{\alpha} N(y)} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial^2}{(\partial y)^2} N(y)}{8\,\alpha tt^{\alpha}} - \frac{t^{2\,\alpha} N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) \frac{\partial}{\partial z} Q(z)^2}{16\,\alpha^2 tt^{\alpha} Q(z)} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} Q(z) \frac{\partial}{\partial y} N(y)^2}{16\,\alpha^2 tt^{\alpha} N(y)}\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = \frac{\alpha N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial}{\partial t} A(t)}{4\,tA(t)^2}$$

$\Psi_1 = 0$

$$\Psi_2 = \frac{\alpha^2 N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z)}{12\, t^2 A(t)} + \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial}{\partial t} A(t)^2}{12\, A(t)^3} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial^2}{(\partial t)^2} A(t)}{12\, A(t)^2} - \frac{N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z)}{12\, t^2 A(t)}$$

$\Psi_3 = 0$

$$\Psi_4 = \frac{\alpha N(y)^{-\frac{1}{\alpha}} Q(z)^{\left(\frac{1}{\alpha}\right)} N(y) Q(z) \frac{\partial}{\partial t} A(t)}{4\, t A(t)^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"